QUANT

Percentages and Profit & Loss

Assignment 02

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Percentage, P & L Online Assignment 02

1. 6 % of the budget of the government is spent on education. If the government increases its budget by Rs.3,000 crores and allocates the entire amount to education, the share of education jumps to 9 %. What is the budget for education in Rs. crores?

1. 6000 2. 5406 3. 5460 4. 8190

2. In an election, half of the electorate voted for Congress, a third of the remaining voted for BJP, a fourth of the remaining voted for Janata and a fifth of the remaining voted for the independent. What % of the total electorate voted in that election?

1. 20% 2. 22.5% 3. 80% 4. 77.5%

3. Khalid writes a number of letters in a day. 36 % of these are to his girl friend. Of the remaining a fifth are business letters and the rest to his family. What % of his letters are to his family?

1. 44% 2. 51.2% 3. 48.8% 4. 56%

4. Of all the Hindi movies that have been made, 11 % were art movies, 33 % were comedy movies, 42 % were action movies and 14 % were tragedy movies. Comedy and action movies expressed as a percentage of art and tragedy movies are -

1. 300% 2. 200% 3. 33.33% 4. 50%

5. Manasi receives a 5 % commission on her sales of exercise equipment and a 6 % commission on her sales of weight training equipment. One month she sold Rs. 79,000 worth of exercise equipment and made a total of Rs. 6,500 in commissions. How much were her total sales of equipment that month?

1. Rs. 121,500 2. Rs. 118,000 3. Rs. 42,500 4. No solution exists

6. When the price of sugar rose by 25 %, a householder reduced his annual consumption by 25 %. His yearly expenditure on sugar then became Rs. 300. What was it before (in Rs.)?

1. 310 2. 350 3. 320 4. 315

7. Fresh mushrooms contain 90% of water (by weight), while dried mushrooms contain 12 % of water. How much dried mushrooms will be obtained from 22 kg of fresh mushrooms?

1. 2.5 kg 2. 3.5 kg 3. 180 kg 4. None of these

8. If the annual salary of a man is Rs.78,000. According to the prevailing tax structure, income up to Rs.50,000 – 0 %. Incremental income is taxed as follows: from 50,001 to 66,000 – 10 % tax; from 66,001 to 75,000 – 15 %; 75,000 – 100,000 - 18.75 %. How much tax does this man pay per month (in Rs.)?

1. 636.35 2. 3,512.5 3. 292.70 4. 725.85

9. Sea fare between Andaman and Calcutta was increased by 3 % as a result of which there was a fall of 2.5 % in sea travel. However, the income of SCI on this sector went up by Rs. 85,000 for the year. What was the original income from this sector prior to the increase (in Rs.)?

1. 2,000,000 2. $2 * 10^5$ 3. $2 * 10^7$ 4. $2 * 10^8$

10. In a constituency of 80,500 voters, 60 % cast their votes. In a straight contest between two candidates, the winner obtained 1,300 votes more than his opponent. By what % of the votes cast, did he beat his opponent in the election?

1. 5.5 % 2. 1.6 % 3. 2.7 % 4. None of these

Answer key & Explanations

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1.	3	Let total budget = x $\therefore \frac{6}{100}x + 3000 = \frac{9}{100}(x + 3000) \Rightarrow x = 9100.$ So budget for education = $\frac{6}{100} \times 9100 = 5460.$

2.	3	$\frac{1}{2}$ voted for Congress, $\frac{1}{3}\times\frac{1}{2}=\frac{1}{6}$ voted for BJP, $\frac{1}{3}\times\frac{1}{4}=\frac{1}{12}$ voted for Janata, $\frac{1}{4}\times\frac{1}{5}=\frac{1}{20}$ voted for Independent, so total = $\frac{1}{2}+\frac{1}{6}+\frac{1}{12}+\frac{1}{20}=\frac{48}{60}$ $\therefore$ required % = $\frac{48}{60}\times 100=80\%$.
3.	2	Suppose he writes 100 letters in a day. He writes 36 letters to his girlfriend. Remaining = 100 – 36 = 64 letters. Business letters = $\frac{1}{5}\times 64=12.8$. So letters to his family = 100 - (36 + 12.8) = 100 - 48.8 = 51.2% because total letters are 100.
4.	1	Total % of art & tragedy movies = 11 + 14 = 25 %. Total % of comedy & action movies = 33 + 42 = 75%. $\therefore$ Required % = $\frac{75}{25}\times 100=300\%$.
5.	1	She gets a commission of 6% on the sales of exercise equipment that becomes 79000 × 6/100 = 3950. The remaining commission is 6500 – 3900 = 2550. This has to be the commission on sales of weight training equipments. Thus weight training equipment sold is 2550 × 100/6 = 42500. The question is asking the sale of equipment so have to include the both i.e. 42500 + 79000 = 121500, thus 1st option.
6.	3	Net effect on expenditure = 1.25 × 0.75 = 0.9375 i.e. decrease of (1 – 0.9375) 100 = 6.25%. Let the earlier expenditure be x $\therefore x - \frac{6.25}{100}x=300 \Rightarrow \frac{93.75}{100}x=300$. Solving this we get x = 320.
7.	1	Fresh mushrooms – 90% water i.e. 10% mushrooms. Dried mushrooms – 12% water i.e. 88% mushrooms. 22 kg of fresh mushrooms contain - 22 × $\frac{10}{100}$ = 2.2 kg mushrooms. Let it contains x kg of dried mushrooms $\therefore x \times \frac{88}{100} = 2.2 \Rightarrow x = 2.5$ kg.
8.	3	Annual salary = Rs. 78000. Total yearly tax = upto 50000 – 0. Next 16000 -10% $\therefore 16000\times\frac{10}{100}=1600$. Next 9000 - 15%. $\Rightarrow \therefore 9000\times\frac{15}{100}=1350$. Next 3000 -18.75% $\therefore 3000\times\frac{18.75}{100}=562.5$ $\therefore$ Total yearly tax = 0 + 1600 + 1350 + 562.5 = 3512.5 $\therefore$ Monthly tax = 3512.5/12 = Rs. 292.70.
9.	3	Net effect = 1.03 × 0.975 = 1.00425 = 0.425% increase $\therefore$ 0.425% of x = 85000 $\frac{425}{1000}\times\frac{1x}{100}=85000$ $\therefore \Rightarrow x = 2\times 10^7$.
10.	3	Total voters = 80500. Votes cast = $\frac{60}{100}\times 80500 = 48300$

		$\therefore$ required % = $\frac{1300}{48300} \times 100 = 2.7\%$.